7.4 Properties of Special Parallelograms

Name _____ Period ____ Date _____

Geometry 7.4 · Student sheet

I can…

- tell whether a statement about rhombuses, rectangles, and squares is always, sometimes, or never true.
- use the diagonals of rhombuses and rectangles to find measures.
- decide whether a quadrilateral in the coordinate plane is a rectangle, rhombus, or square.

Vocabulary · fill in the missing word

Rhombus — a _____ with four congruent sides

Rectangle — a _____ with four right angles

Square — a _____ with four congruent sides and four right angles

Part 1 · Property chart

Write **yes** or **no** in each box.

Property	Parallelogram	Rectangle	Rhombus	Square
Opposite sides parallel and congruent				
All four sides congruent				
All four angles right				
Diagonals bisect each other				
Diagonals congruent				
Diagonals perpendicular				
Diagonals bisect opposite angles				

Family tree. Write quadrilateral, parallelogram, rectangle, rhombus, and square in the boxes.

Part 2 · Notes and examples (fill in each note, then work the example)

Rhombus: A **rhombus** is a parallelogram with four _____.

Rectangle: A **rectangle** is a parallelogram with four _____.

Square: A **square** is a parallelogram with four _____ and four _____.

Example 1

Decide whether each statement is always, sometimes, or never true.

a. A rectangle is a square.

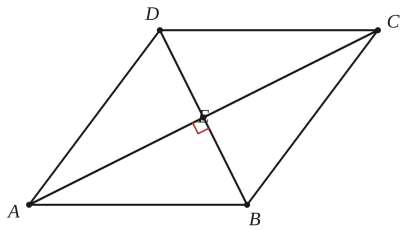
b. A square is a rhombus.

c. A rhombus has congruent diagonals.

Rhombus Diagonals and Opposite Angles Theorems: The diagonals of a rhombus are _____, and each diagonal _____ a pair of opposite angles.

Example 2

ABCD is a rhombus with diagonals meeting at E, and m∠ABC = 70°. Find m∠ABD and m∠AEB.



Rectangle Diagonals Theorem: The diagonals of a rectangle are _____.

Example 3

In rectangle ABCD, AC = 4x − 3 and BD = 2x + 9. Find x and AC.

Example 4

Decide whether the parallelogram with vertices (−2, 1), (1, 3), (4, 1), and (1, −1) is a rectangle, rhombus, or square. Explain.

Part 3 · Partner practice

Partner 1

In rhombus JKLM, JK = 3x + 5 and KL = 5x − 7. Find x and the length of each side.

Partner 2

In rectangle JKLM, JL = 5y − 8 and KM = 3y + 6. Find y and JL.